

Alexander Vega-Hernandez

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EDUCATION

University of Michigan–Ann Arbor

Bachelor of Science in Computer Science

Ann Arbor, MI

August 2025

Relevant Coursework: Distributed Systems, Algorithms and Data Structures, Databases, Web Systems, Computer Security, Computer Architecture, Software Engineering

TECHNICAL SKILLS

Languages: C++, C, Python, Go, JavaScript, SQL, HTML/CSS

Frameworks/Tools: React, Flask, Node.js, Docker, Git, Bash | **Databases:** PostgreSQL, SQLite, MySQL, MongoDB | **Cloud:** AWS

EXPERIENCE

R&D Software Engineer Intern

May 2026 – August 2026

Dassault Systèmes | C++, 3DExperience, HTML

Royal Oak, MI

- Built debug tooling for a multi-robot path planning system, automating voxel pose updates and adding a route feasibility validator on existing planner data; adopted by teammates for day-to-day debugging.
- Reimplemented a BFS-based pathfinding routine with Dijkstra's algorithm, which a teammate adopted in his downstream component and reported faster runs; built real-time UI tooling for path inspection.
- Researched and built a 3D confidence map and medial axis extraction tool for spot-welding workspaces using a distance transform and the Hamilton-Jacobi PDE, laying the groundwork for the team's planned Fast Marching Tree (FMT) integration to give robots full 3D spatial awareness.

PROJECTS

Sharded Key-Value Store with Paxos Consensus | Go, Distributed Systems

- Engineered a fault-tolerant key-value store on custom Paxos, passing all 38 fault scenarios across 3-replica storage and a 16-shard controller, including network partitions, unreliable RPCs, and minimal-transfer rebalancing.

AI-Powered Mental Health Web App | React, Python, Flask, SQLite, Gemini API

- Delivered full-stack features for a 4-person team's Gemini journaling app: 8 of 11 Flask REST endpoints, React analytics charts, and mood-specific AI prompts with output guardrails.

Distributed MapReduce Framework | Python, Distributed Systems

- Implemented a Manager/Worker MapReduce runtime with TCP task dispatch and UDP heartbeats, passing all 18 functional tests, including dead-worker detection, task reassignment, and a 28% faster map stage.

Search Engine | Python, Flask, SQL, Web Systems

- Developed a search engine over 3,062 Wikipedia pages (1.1 GB), using a 6-stage MapReduce pipeline to build a 3-segment inverted index; ranked results by tf-idf and PageRank via Flask REST index servers queried concurrently.

LEADERSHIP

Project Manager & Team Captain

June 2021 – August 2021

Henry Ford Learning Institute Summer Capstone

Detroit, MI

- Led a 5-person team designing a college guidance app for Latinx students; refined layout from user research, ran cross-functional design reviews, and pitched final blueprints to HFLI community leaders.

ADDITIONAL INFORMATION

Languages: English, Spanish (Bilingual)

Interests: R&D, Software Maintenance and Scalability, Open Source Contribution, UI/UX Design, CAD